

# Knowledge Base (KB) Library

Reusable knowledge for diagnostics and Root Cause Analysis (RCA) throughout the Minimum Viable Product (MVP).

OBJECTIVE, SCOPE & DESIGN

Manage each KB as reusable, versioned knowledge with a named consumer and a clear purpose.  
Library = workspace • KB = governed collection • Version = a saved edition • Entry = a rule or concept  
Source Documents supply evidence. Users can build, test and promote KBs, or change settings for one run.

DIAGNOSTIC FRAMEWORK CONSUMPTION

Design focus: T2-D06 Row completeness • T2-D11 Directionality  
T2-D08 Value semantics (detailed design to follow)

FUTURE LIBRARY ADOPTION — LIST ONLY

T1-D02 Target separation  
T2-D04 Cross-field business rules  
T3-D12 Label consistency  
T4-D14 Population stability  
T5-D17 Resolution / maturity by vintage  
T6-D20 Artificial Intelligence (AI) proposals

Deferred framework coverage:  
T1-D01 Look-ahead; D03 Categorical leakage  
T2-D05 Derivation; D07 Disguised missingness  
T2-D09 Outliers; D10 Boundary / pile-up  
T3-D13 Target-rate breaks  
T4-D15 Distribution distance; D16 Segment coverage  
T5-D18 Resolved vs all; D19 Seasoning profile

RCA & ISSUE RESOLUTION

Use applicable KB rules, definitions and source evidence to investigate an issue.

Finding → reviewed issue → RCA → confirmed cause or unresolved outcome.

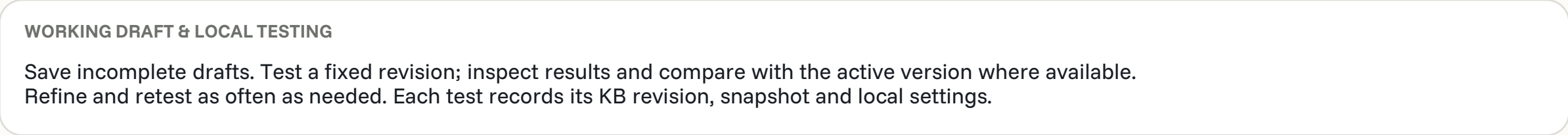
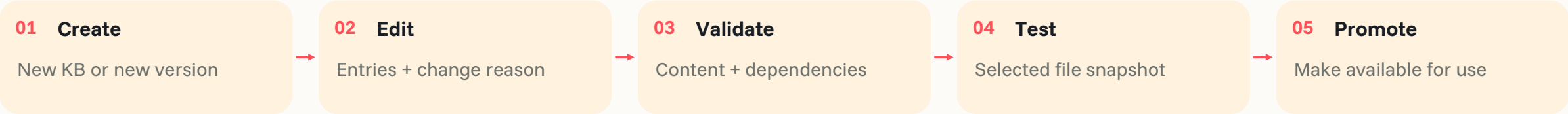
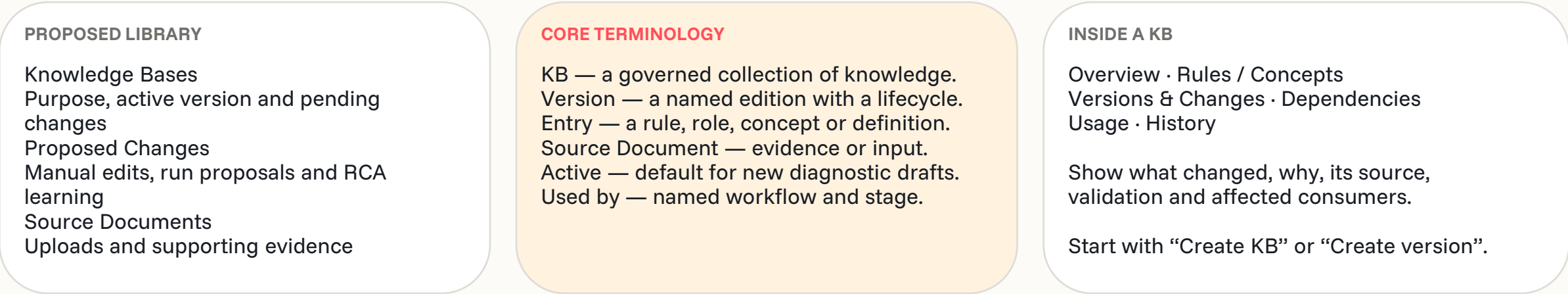
Reusable learning → proposed KB change → validate and promote → future use.

Confirm a correction by reassessing data.

Future enhancement: Role-Based Access Control (RBAC) and separation of duties. MVP records actors and decisions; self-review is allowed.

# The Library and the working journey

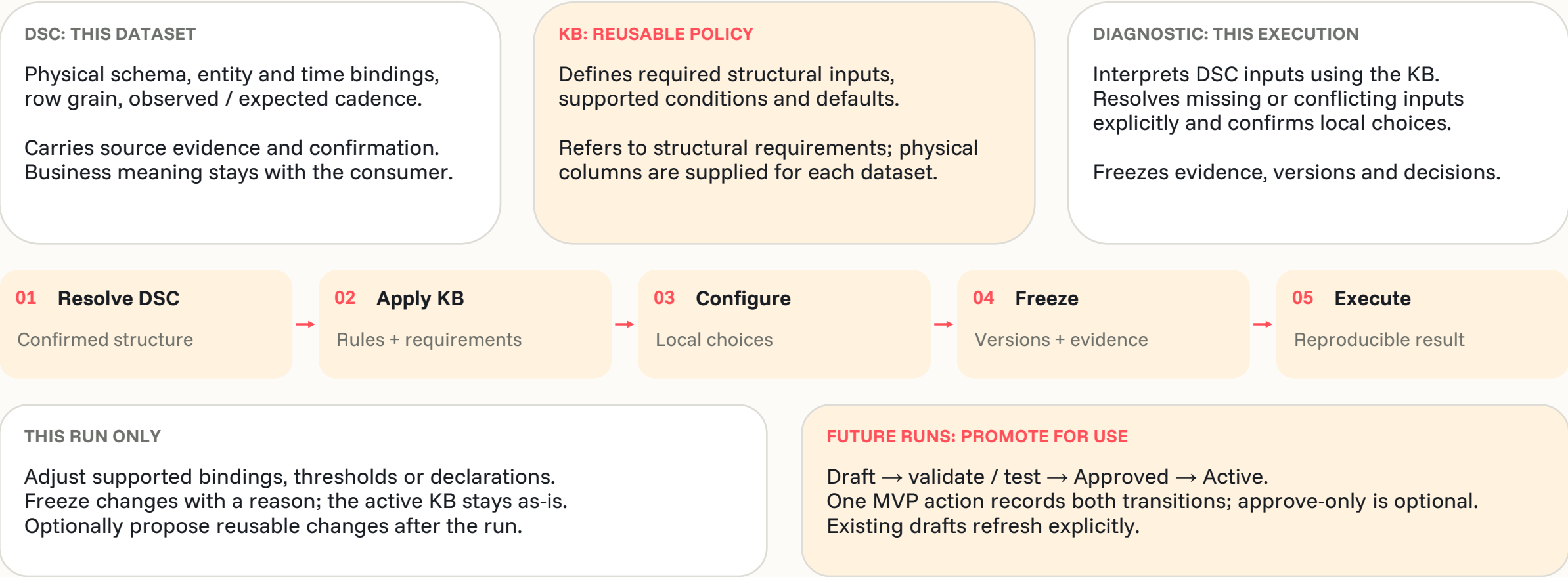
Create a KB from scratch or refine an existing version through repeated edit, validate and test cycles.



Draft tests are labelled clearly. Promotion makes a validated version available to future diagnostic drafts.

# Structure, local choices and promotion

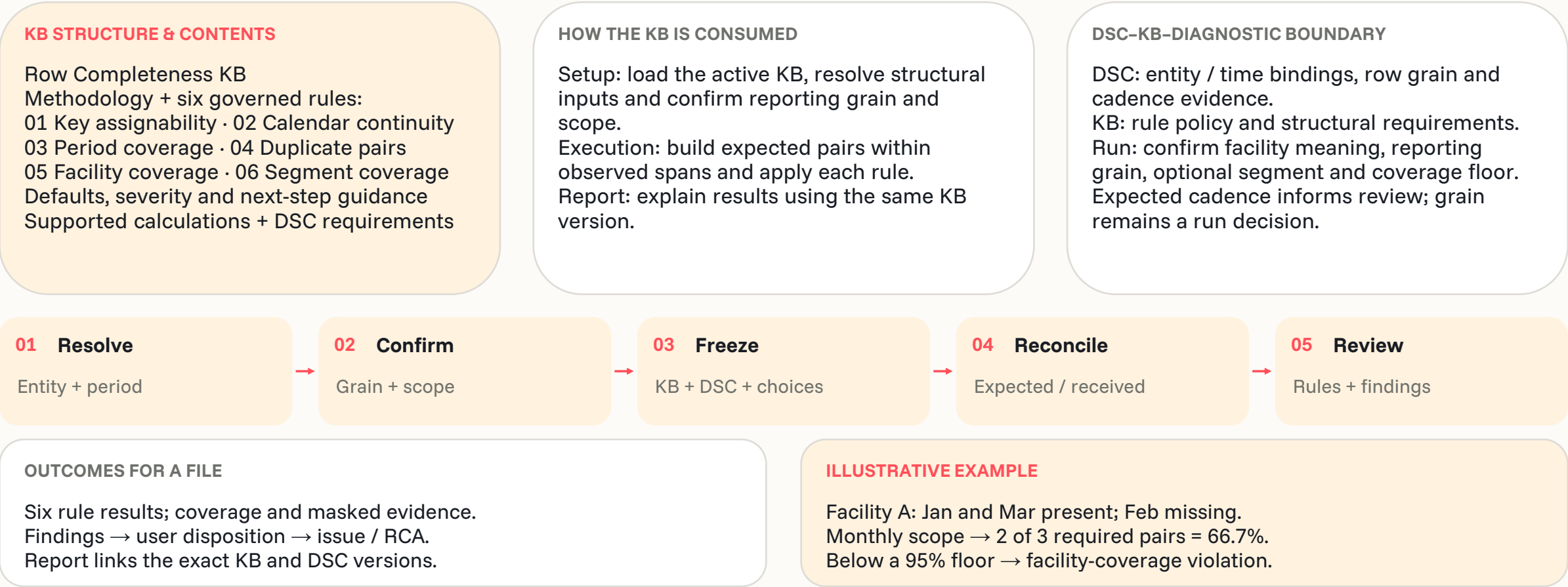
Dataset Structure Context (DSC) provides evidence about the structure of one immutable dataset snapshot.



A new active version supersedes the old one. Withdrawal has no automatic fallback. Frozen runs retain their versions.

# T2-D06 | Row completeness

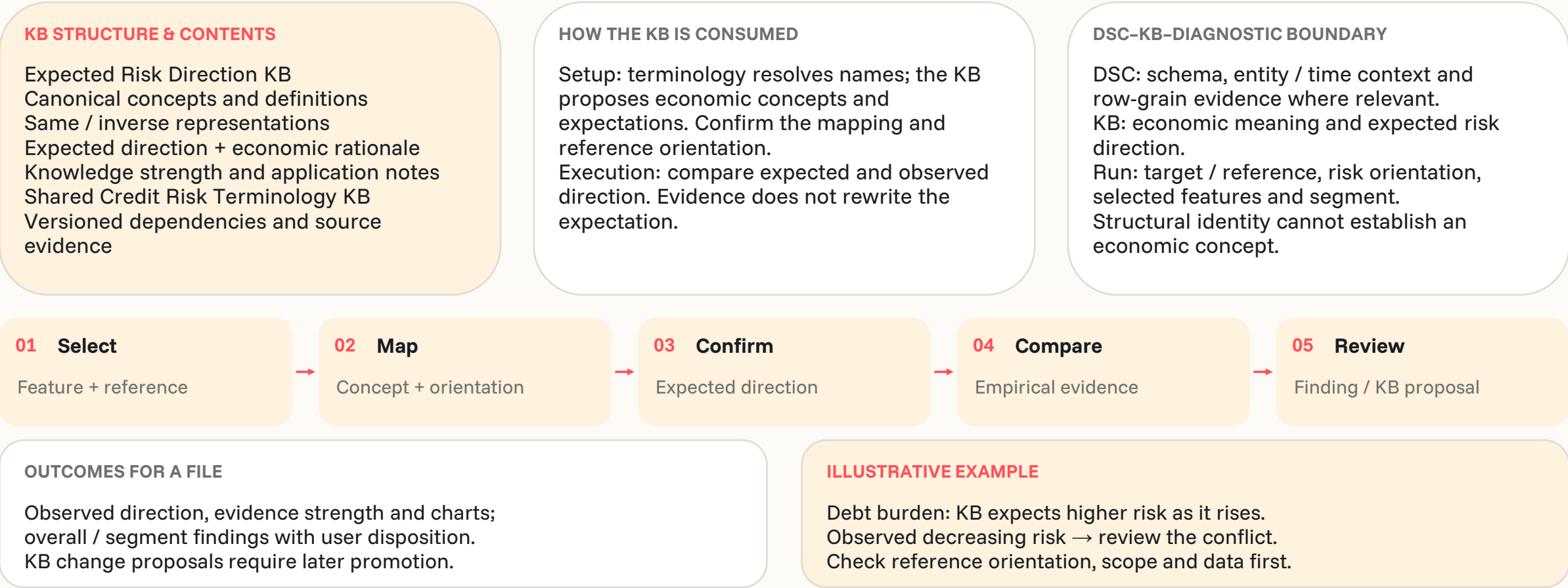
Objective: assess required facility-period rows within observed spans and identify gaps, invalid keys and duplicates.



Scope: observed spans cannot detect wholly absent facilities or missing rows beyond observed endpoints.

# T2-D11 | Directional monotonic consistency

Objective: compare a feature’s expected relationship to risk with observed behavior against a confirmed reference.



A local expected-direction change affects one run. Reusable changes enter a KB draft before future adoption.

# T2-D08 | Value semantics

Objective: explain values that are not final, unexpectedly frozen or inapplicable before interpreting data-quality signals.

**KB STRUCTURE & CONTENTS**

Value Semantics KB — initial framing  
Semantic roles, definitions and cardinality  
Classification entries and prerequisites  
Typed business declarations  
Precedence, safeguards and exclusions  
Supported calculations + DSC requirements  
Shared terminology dependency

**HOW THE KB IS CONSUMED**

Setup: match fields to semantic roles;  
confirm bindings and supply known  
declarations.  
Execution: route applicable entries and  
resolve cell classifications.  
Report: show tags, coverage and evidence  
for analytical treatment.

**DSC-KB-DIAGNOSTIC BOUNDARY**

DSC: schema, entity / time structure and  
observed cadence where supported.  
KB: business roles, rule conditions and  
precedence.  
Run: sentinel definitions, observation  
windows, process states and applicability.  
Structural evidence alone does not  
establish value meaning.

**01 Understand**

Purpose + scope

**02 Bind**

Confirm field roles

**03 Declare**

Known business facts

**04 Classify**

Freeze + evaluate

**05 Review**

Tags + coverage

**OUTCOMES FOR A FILE**

CENSORED · STALE\_FROZEN · NOT\_APPLICABLE  
NO\_TAG · UNCLASSIFIED · UNSCOPED coverage  
Grouped evidence; user-controlled issue / RCA.

**ILLUSTRATIVE EXAMPLE**

A 12-month outcome has only 6 months of  
confirmed observation coverage → CENSORED.  
A populated value may still be unfinished.

Initial framing only: D08 end goal, tag vocabulary, rule content and redesign scope remain for the next design review.